

FinControl AI

AI-powered reconciliation, financial controls, exception-risk analytics, and Power BI-ready reporting

PROJECT STATUS	PUBLISHED REPOSITORY
Complete	github.com/cosmosandyou/fincontrol-ai

Executive summary

FinControl AI is a portfolio-ready finance-controls demonstration that turns synthetic payment, bank-statement, general-ledger, customer-account, foreign-exchange, and control-rule data into explainable reconciliation results. It identifies breaks, assigns risk, detects unusual patterns, and produces curated data products for Streamlit and Power BI.

636	14	52	39
matched transactions	reconciliation breaks	exceptions generated	anomalies flagged

What the solution delivers

Capability	Outcome
Reconciliation	Matches payment records to bank statements and ledger postings, separating matched, mismatched, and unmatched activity.
Control testing	Runs deterministic checks for missing settlements, duplicate bank records, delayed settlement, FX, ledger, and classification issues.
Risk and anomaly signals	Combines control failures with explainable risk scores and Isolation Forest anomaly detection.
Decision support	Provides a Streamlit dashboard with executive, investigator, exception, anomaly, and regulatory-readiness views.
BI handoff	Exports audited CSV outputs for Power BI reporting and refresh workflows.

Architecture and workflow

The project is intentionally compact and reproducible. A deterministic generator produces synthetic source-system data. The pipeline reconciles source records, executes control checks, calculates risk and anomalies, and exports audit-ready datasets. The Streamlit application and Power BI report consume these same curated outputs.

Step	Layer	Detail
1	Synthetic source data	Payments, bank statements, general ledger, customer accounts, FX rates, and control rules.
2	Reconciliation pipeline	Transaction matching, mismatch identification, and break classification.
3	Controls and risk	Deterministic control failures, risk levels, recommended action, and anomaly signals.
4	Curated outputs	Exceptions, reconciled transactions, reconciliation summary, and control summary CSVs.
5	Decision interfaces	Streamlit dashboard and Power BI import model for operations, audit, and management.

Generated reconciliation results

Status	Transactions	Value (EUR)
Matched	636	224,972.70
Mismatch	6	803.63
Unmatched	8	3,080.21

Exception and control findings

The generated output contains 52 exceptions across nine identified categories. The highest-priority findings include missing bank settlements and missing regulatory classifications. Unusual transaction patterns account for the largest observed value at risk, making anomaly investigation a practical triage use case.

Finding	Risk level	Count	Value at risk (EUR)
Missing bank settlement	High	5	1,954.85
Missing regulatory classification	High	1	1,282.57
Unusual transaction pattern	Medium	18	26,315.19
Unusual transaction pattern	Low	12	9,874.34
Amount mismatch	Medium	6	803.63

Dashboard and Power BI reporting

The Streamlit interface is structured for both senior stakeholders and operational investigators. It exposes executive risk indicators, reconciliation status, filtered exception tables, anomaly analysis, regulatory readiness, and deterministic recommended actions. The Power BI guide provides measures and a six-page report design.

View	Purpose
Executive Risk Overview	KPIs, risk distribution, trend, and priority exception table.
Reconciliation Monitor	Matched/unmatched/mismatch analysis and break drill-through.
Exception Explorer	Provider, risk, currency, and type filters with audit notes.
Anomaly Detection	Amount versus settlement-days analysis and anomaly filtering.
Regulatory Readiness	Classification completeness and control-failure monitoring.
AI Investigator	Evidence-led investigation support and recommended next action.

Power BI model and measures

Load **exceptions.csv**, **reconciled_transactions.csv**, **reconciliation_summary.csv**, and **control_summary.csv**. Relate exceptions (many) to reconciled transactions (one) using `transaction_id`. Useful starter measures include Total Exceptions, High Risk Breaks, Value at Risk, Unresolved Issues, and Control Pass Rate.

Repository and operating instructions

The source code and documentation are published at github.com/cosmosandyou/fincontrol-ai. To run locally: install requirements, generate synthetic data, run the pipeline, then start Streamlit. The project uses synthetic data only and is intended as a demonstrator, not a production financial-control framework.

Command
<code>python -m pip install -r requirements.txt</code>
<code>python -m src.generate_data</code>
<code>python -m src.pipeline</code>
<code>streamlit run app.py</code>

Recommended next step

Deploy the Streamlit app to Streamlit Community Cloud or another Python-capable host and add its URL to the repository README. For a production adaptation, replace synthetic sources with approved connectors, protect secrets, add role-based access, retain evidence, and require human review for investigative conclusions.